

Optimiser

A broadcast-style loudness chain: AGC, four-band compression, clipper and limiter.



Overview

Optimiser is a loudness processor in the spirit of the classic broadcast processors that made FM stations sound loud, dense and consistent. The signal runs through a chain: a slow AGC that rides the level, a phase rotator, a four-band compressor, an HF limiter, a soft clipper and finally a brickwall limiter.

The four bands are split at 150 Hz, 1.2 kHz and 6 kHz with a phase-coherent crossover that sums back flat. DRIVE pushes all four compressors harder; BASS, MID, PRESENCE and AIR set each band's output level, which is where the station's 'sound' comes from. The bands are linked by COUPLING, so the tonal balance holds while they work.

MIX blends the processed sound with the unprocessed one (passed through the same crossover, so the two stay in phase) for parallel processing.

Reach for it for

- Making a master or a stem loud and dense
- Radio-style, 'on air' processing of a mix
- Evening out speech and podcasts
- Parallel punch on drums or a mix bus
- Lo-fi AM-radio effects

Controls

Display

Bar meters	One bar per stage. AGC: the gain the AGC is adding (up from the centre line) or taking away (down), up to 12 dB. BASS, MID, PRES, AIR: each band's gain reduction. HF: the HF limiter. CLIP: how hard the clipper is working. LIMIT: the final limiter. The scale runs from 0 at the top to 20 dB at the bottom.
INPUT	A VU meter showing the input level (0 VU = -18 dBFS).

Header

SLOW / MED / FAST	How quickly the four band compressors recover. SLOW is open and natural; FAST gives a denser, more obviously processed sound.
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PROCESS

AGC	Switches the AGC on: a slow wideband gain rider that brings the average level towards a steady -18 dBFS, adding or taking away up to 12 dB. It rises at 2 dB per second and falls at 6 dB per second.
GATE	Only with AGC on (on by default). Freezes the AGC when the signal falls below -50 dBFS, so fades and pauses are not pumped back up.

DRIVE	Pushes all four band compressors harder, 0 to +18 dB. More DRIVE means more compression and a denser sound.
BASS	The level of the low band (below 150 Hz), -6 to +6 dB.
MID	The level of the low-mid band (150 Hz - 1.2 kHz), -6 to +6 dB.
PRESENCE	The level of the upper-mid band (1.2 - 6 kHz), -6 to +6 dB.
AIR	The level of the top band (above 6 kHz), -6 to +6 dB.
CLIP	Drives the soft clipper ahead of the final limiter, by up to 9 dB. It rounds off the peaks for extra loudness and bite.

SHAPE

ROTATOR	On by default. Four all-pass filters at 200 Hz turn lopsided peaks (typical of speech) into a more symmetrical wave, so the clipper and limiter get more loudness for the same distortion. It changes phase, not tone.
FIZZ	The broadcast exciter: adds only new harmonics from everything above 3 kHz, shaped around 8 kHz and held about 26 dB below the programme, blended in parallel. A little sparkle on top.
COUPLING	How tightly the four bands are linked, 0 to 100% (50% by default). At 100% they move as one; at 0 they work fully independently. In between, each band's gain reduction can stray only so far from the average of the four, so the tonal balance holds.
HF LIMIT	A fast limiter on the top band, 0 to 100% (40% by default). It holds the treble at a steady, pinned level - the bright, even broadcast top end - and stops the highs from hitting the clipper first. Fully left is off.

DRIFT

DRIFT	See Drift below.
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IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB. A single click on the knob switches GUARD on or off (off by default); its light at the knob's upper right is lit while GUARD is on and glows brighter while it is catching peaks. GUARD gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage, LIMIT by default. OFF: nothing. CLIP: a soft clipper that clips the peaks at the ceiling (-0.3 dBFS by default); THRESH does nothing in this mode. LIMIT: a lookahead brickwall limiter (3 ms latency, compensated by REAPER) that holds the peaks at the ceiling. The CLIP key flashes red while the output is being clipped.

MASTER

MIX	Blends the processed signal with the unprocessed one, 0 to 100%. Below 100% you get parallel processing: the dense, loud sound under the natural dynamics.
OUTPUT	The output level, -24 to +12 dB.
THRESH	With the LIMIT stage: pull it down (to -24 dB) to push the whole chain harder into the limiter's ceiling - louder, with the peaks held. The clipper starts at the same point. With CLIP or OFF it reads LIMIT OFF.
GR	The limiter's gain reduction, 12 lights of 1 dB each, lit from the top down.
OUT meters	Output level, left and right. Click them to reset the peak hold.

The header

Theme picker	Changes the panel's colour theme.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
HW	Switches between the hardware-style panel and the flat look.
PRESETS	Opens the list of factory presets (or click the preset display). The ◀ / ▶ keys step through them. Choosing one sets the release speed, AGC and gate, DRIVE, the four band levels, CLIP and MIX; the SHAPE section, drift, OUTPUT and the input / output settings are left as they are. The last entry resets everything to the defaults.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Drift

DRIFT (0 - 100%, 50% by default) makes the processor breathe slightly, like an unsteady power supply: the drive into the bands wanders by a fraction of a dB and the bands' attack and release times waver. 50% is the standard amount, 100% twice that, 0 none; the wander also speeds up as you turn it up.

Presets

Gentle FM	Slow release, AGC on, light drive with a touch of bass, presence and air; a little clipping.
Rock FM	Medium release, more drive, a lift in the lows and highs, moderate clipping.
Pop Loud	Fast release, heavy drive and a bright, bassy balance with plenty of clipping.
Talk Radio	Medium release, bass cut and a mid and presence lift for clear speech.

Classical	Slow release, very light drive and no clipping, to keep the dynamics.
Dance Smash	Fast release, AGC gate off, very heavy drive, big bass and air, hard clipping.
Vintage AM	Heavy drive, lifted bass, mids and presence with the air cut by 6 dB, hard clipping: the old AM-radio sound.
Master Polish	AGC off, slow release, light drive with a touch of bass, presence and air.
Podcast Voice	Medium release, bass cut and a presence lift for spoken word.
Parallel Punch	Fast, heavily driven and clipped, blended in at 40% mix.
Reset all to defaults	Every control back to its starting value.

Tips

- Set the sound with DRIVE and the four band levels first, then use THRESH to set the final loudness.
- Watch the band bars: if one band's reduction is far bigger than the others, lower that band's level or raise COUPLING.
- If the top end gets harsh as you push harder, raise HF LIMIT before you reach for AIR.
- Keep the AGC for programmes with uneven levels (speech, live recordings); turn it off for a finished master you have already levelled.
- For drums or a mix bus, try MIX at 30-50% for parallel density without losing the transients.

File: gfx-optimiser.jsfx